

3 The photograph shows one variety of wheat and the grains it produces.



(Source: © Li Ding/Alamy Stock Photo)

A scientist observed that the germination of wheat grains was affected by the presence of arsenic ions in the soil.

Groups of 30 grains were placed on filter paper in Petri dishes. Each Petri dish had 0.5 cm³ of water with a different concentration of arsenic ions added.

The Petri dishes were incubated at 23 °C for four days. The percentage of seeds that germinated was recorded.

The procedure was repeated a number of times.

The table shows the results of this investigation.

Concentration of arsenic ions / mg dm^{-3}	Mean percentage of seeds germinated after 4 days %
0.0	100
0.5	84 ± 2
1.0	85 ± 2
2.0	81 ± 6
4.0	77 ± 8
8.0	65 ± 14
16.0	66 ± 2

- (a) (i) Identify **one** risk associated with this investigation and how you could reduce this risk.

(2)

Risk

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How to reduce this risk

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- (ii) State a suitable null hypothesis for this investigation.

(1)

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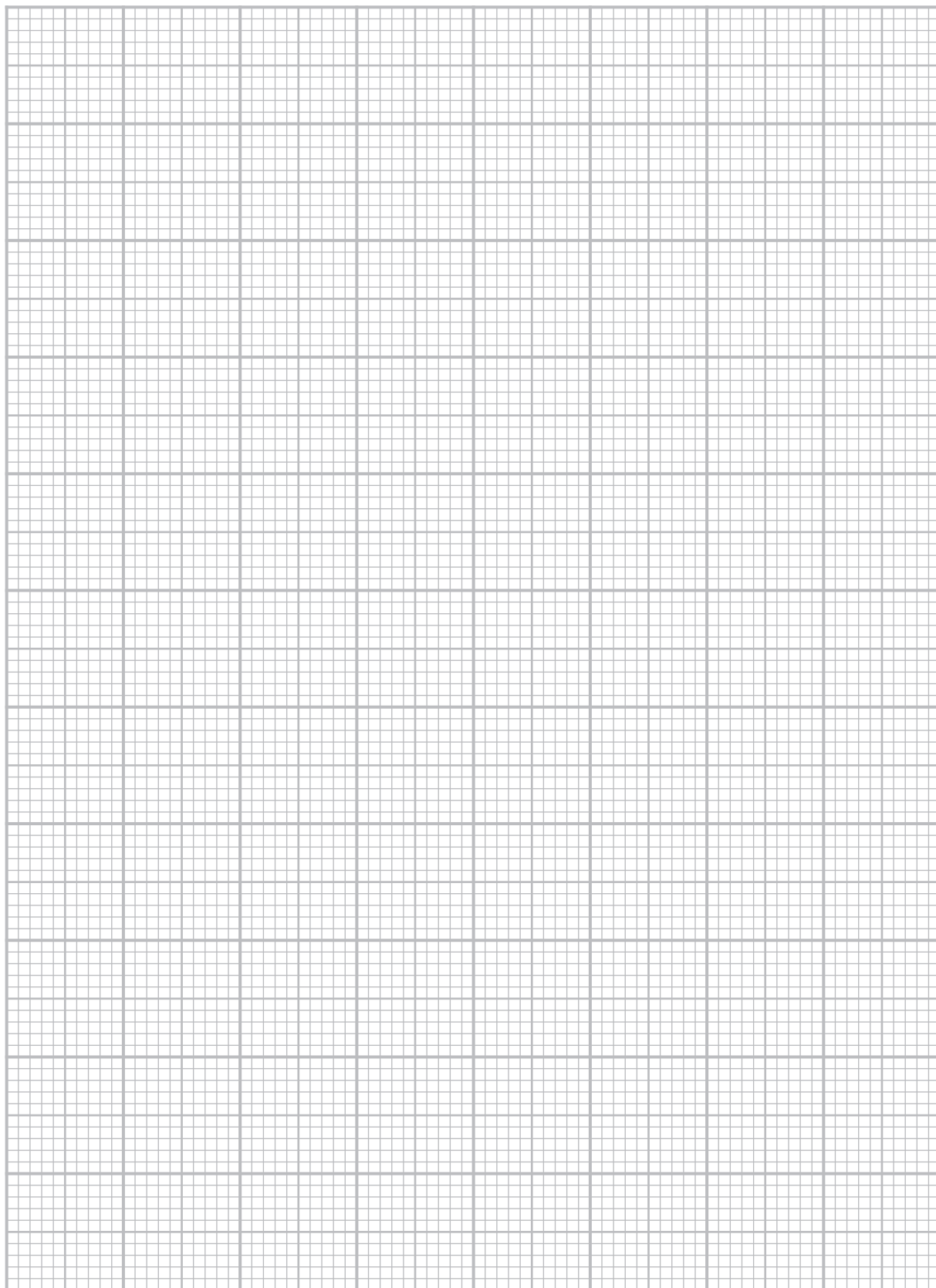
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(iii) Plot a suitable graph to show the effect of the concentration of arsenic ions on the percentage germination of wheat grains.

(3)



(b) The scientist used a statistical test to analyse the data.

To calculate the correlation coefficient, the scientist produced this table.

Concentration of arsenic ions / mg dm ⁻³ (a)	Mean percentage of grains germinated after 4 days (%) (b)	Rank (a)	Rank (b)	<i>d</i>	<i>d</i> ²
0.0	100	1	1	0	
0.5	84	2	3	-1	
1.0	85	3	2	1	
2.0	81	4	4	0	
4.0	77	5	5	0	
8.0	65	6	7	-1	
16.0	66	7	6	1	

(i) Calculate the correlation coefficient (r_s) using the formula:

(2)

$$r_s = 1 - \frac{6\sum d^2}{n(n^2 - 1)}$$

Where:

$\sum$ = the sum of

d = the difference between each pair of ranks

n = the size of the sample (number of pairs of values)

Answer

(ii) The table shows some critical values for this statistical test.

Number of pairs of values	Level of significance (p)		
	0.10	0.05	0.01
4	1.000	–	–
5	0.900	1.000	–
6	0.829	0.886	1.000
7	0.714	0.786	0.929
8	0.643	0.738	0.881
9	0.600	0.700	0.833
10	0.564	0.648	0.794

Comment on the conclusion that can be drawn from this investigation.

Use your graph, your calculated r_s value and the table of critical values to support your answer.

(3)

[illegible]

- (c) (i) In breeding experiments, scientists select varieties of wheat with different characteristics.

Explain why some varieties of wheat have different tolerances to arsenic ions.

(2)

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- (ii) Explain how this investigation could be extended to study the effect of arsenic ions on the growth of wheat plants from germinating wheat grains.

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(Total for Question 3 = 16 marks)